(i) Total sum = 36, Total sum of squares = 204.
Each subset's sum= 18, sum of squares = 102.

$$\boxed{A = \{1, 4, 6, 7\}, \qquad B = \{2, 3, 5, 8\}.}$$

(ii)

$$\text{LHS} = \sum a_k^3 + \sum \left(c^3 + 3c^2 b_k + 3c b_k^2 + b_k^3\right) = \sum a_k^3 + \sum b_k^3 + mc^3 + 3c^2 \sum b_k + 3c \sum b_k^2$$

$$\because \sum a_k = \sum b_k \text{ and } \sum a_k^2 = \sum b_k^2$$

$$\begin{aligned} \therefore \text{LHS} &= \sum a_k^3 + \sum b_k^3 + mc^3 + 3c^2 \sum a_k + 3c \sum a_k^2 \\ &= \sum_{k=1}^{m} b_k^3 + \sum_{k=1}^{m} (c + a_k)^3 \end{aligned}$$

(iii) From part (i), let

$$A' = \{1, 4, 6, 7\}, \qquad B' = \{2, 3, 5, 8\}.$$

Then $\sum a'_k = \sum b'_k$ and $\sum {a'_k}^2 = \sum {b'_k}^2$.
Choosing $c = 8$

$$\boxed{A = \{1, 4, 6, 7, 10, 11, 13, 16\}, \qquad B = \{2, 3, 5, 8, 9, 12, 14, 15\}.}$$

Now

$$\sum_{x \in A} x = \sum a_k + \sum (8 + b_k), \qquad \sum_{x \in B} x = \sum b_k + \sum (8 + a_k),$$

Since $\sum a'_k = \sum b'_k, \quad \sum a_k = \sum b_k$

$$\begin{aligned} \sum a_k^2 + \sum (c + b_k)^2 &= \sum a_k^2 + \sum (c^2 + 2c b_k + b_k^2) \\ &= \sum a_k^2 + mc^2 + 2c \sum b_k + \sum b_k^2 \\ &= \sum b_k^2 + mc^2 + 2c \sum a_k + \sum a_k^2 \\ &= \sum b_k^2 + \sum (c + a_k)^2 \end{aligned}$$

So, the sum of squares of A and B are equal.
By part (ii),

$$\sum a_k^3 + \sum (8 + b_k)^3 = \sum b_k^3 + \sum (8 + a_k)^3,$$

Thus, the sum of cubes of A and B are equal.

(iv) Since $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{2, 6, 7, 8, 10\}$ form a balanced partition of $\{1, 2, \ldots, 11\}$, it follows that

$$\{1, 3, 4, 5, 9, 11\} \quad \text{and} \quad \{0, 2, 6, 7, 8, 10\}$$

form a balanced partition of $\{0, 1, 2, \ldots, 11\}$, and each subset contains 6 elements.
Hence

$$\{n+1, n+3, n+4, n+5, n+9, n+11\}$$

and

$$\{n, n+2, n+6, n+7, n+8, n+10\}$$

form a balanced partition of

$$\{n, n+1, n+2, \ldots, n+11\}.$$

Hence

$$\boxed{C = \{(n+1)^2, (n+3)^2, (n+4)^2, (n+5)^2, (n+9)^2, (n+11)^2\}}$$

and

$$\boxed{D = \{n^2, (n+2)^2, (n+6)^2, (n+7)^2, (n+8)^2, (n+10)^2\}}$$

is a pair of sets with the required property.

This solution was submitted by **Jon**.

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